



HAMZA EL MAZOUZI

Data Engineer

✉ hamzaelmazouzi@gmail.com

☎ (+212) 0684653903

📍 Tangier, Morocco

PROFESSIONAL EXPERIENCE

Data Engineer

Moroccan Ministry of Interior

March 2021- Present – Tangier

- Designed and implemented a data pipeline using PySpark, Java, and Selenium to extract, transform, and load data from Moroccan government websites, to manage the Morocco general election of 2021. Developed a data warehouse using PostgreSQL to store and manage the extracted data. Performed data cleaning and transformation using PySpark, and loaded the cleaned data into the PostgreSQL data warehouse for further analysis and reporting.
- Designed error handling mechanisms and monitoring functionality in the data pipeline to ensure the reliability and accuracy of the data. Ensured the pipeline was scalable and maintainable by designing it with modularity and extensibility in mind.
- Developed impactful data visualization dashboards using Tableau and Power BI to support data-driven decision-making in various development projects in the region. Designed visually appealing and informative dashboards for tracking the progress of these projects, which allowed stakeholders to quickly gain insights into project performance and identify areas that needed attention.
- Designed a scalable and reliable data architecture to support data visualization dashboards. Built ETL pipelines using a combination of technologies, such as Python, SQL, and Apache Airflow, to extract data from various sources. Implemented data validation checks using tools such as Apache Nifi to ensure data quality. Designed a centralized data warehouse using PostgreSQL to store and manage the cleaned and transformed data.

Data Engineer

ALTEN DELIVERY CENTER

August 2019- February 2021 – Fez

- Managed the data pipeline for the manufacturing plant's production process for PSA Group. Implemented a data pipeline using Kafka on Red Hat Linux to effectively manage and analyze the large volume of data generated by the manufacturing plant.
- Utilized Airflow to automate and schedule the data pipeline. Created Airflow DAGs to trigger the Kafka producers and consumers at specific intervals and to manage the ETL processes that transformed the data in the data lake stored in Amazon S3. Additionally, Airflow was used to orchestrate the data loading process from S3 to Amazon Redshift and to schedule the ETL jobs that transformed the data in Redshift. This improved the efficiency and reliability of the data processing and analysis tasks, resulting in faster and more accurate insights for the business users.
- Developed and maintained modules for a platform that processed vehicular testing data using Java. Collaborated with a team to design and implement new features, debugged issues, and ensured proper performance. Used Tortoise SVN to manage version control.
- Managed data quality and integrity through data validation, data cleansing, and data transformation.

EDUCATION

Master's Degree in Big Data

Euro-Mediterranean University in Fez

September 2017- July 2019

SUMMARY

Data Engineer with over 4 years of experience in designing, building, and maintaining data pipelines, ETL processes, and data warehousing solutions. Proficient in programming languages such as Python, SQL, and Java. Strong background in data modeling, data governance, and data quality. Familiarity with big data technologies such as Hadoop, Spark, and Kafka.

SKILLS

- Programming Languages: Python, Java, Scala, SQL
- Big Data Technologies: Kafka, Spark, Hadoop
- Databases: PostgreSQL, MySQL, MongoDB
- Cloud Platforms: AWS (Redshift, S3)
- ETL Processes: Data modeling, database design, data integration, data warehousing
- Data Quality Management: Data cleansing, data validation, data transformation
- Data Visualization Tools: Tableau, PowerBI
- OS: Windows, Linux (Red Hat, Ubuntu)